

Task 1 – CSV File: Student Marks Record

Problem Statement

Develop a Python program to collect the details of 10 students and store them in a CSV file named student_marks.csv.

The file should contain:

- Register Number
- Student Name
- Department
- Python Marks
- Data Science Marks

CODE:

```
import csv
```

```
with open("student_marks.csv", "w", newline="") as file:
```

```
    writer = csv.writer(file)
```

```
    writer.writerow(["Register No", "Student Name", "Department", "Python  
Marks", "Data Science Marks"])
```

```
    for i in range(3):
```

```
        reg = input("Register No: ")
```

```
        name = input("Student Name: ")
```

```
        dept = input("Department: ")
```

```
        py = input("Python Marks: ")
```

```
        ds = input("Data Science Marks: ")
```

```
        writer.writerow([reg, name, dept, py, ds])
```

```
print("Student records saved successfully.")
```

```
Register No: 192324092
Student Name: sabarish \
Department: aids
Python Marks: 99
Data Science Marks: 90
Register No: 192324093
Student Name: anto
Department: aids
Python Marks: 99
Data Science Marks: 90
Register No: 192324094
Student Name: ajmal
Department: aids
Python Marks: 90
Data Science Marks: 90
Student records saved successfully.
```

Task 2 – CSV File: Employee Database

Problem Statement

Develop a Python program to collect employee information and store it in a CSV file named employees.csv.

The file should contain:

- Employee ID
- Employee Name
- Department
- Designation
- Salary

Store details of at least 5 employees.

```
import csv

with open("employees.csv", "w", newline="") as file:
    writer = csv.writer(file)

    writer.writerow(["Employee ID", "Employee Name", "Department", "Designation",
                    "Salary"])

    for i in range(2):
        eid = input("Employee ID: ")
        name = input("Employee Name: ")
        dept = input("Department: ")
        des = input("Designation: ")
        sal = input("Salary: ")

        writer.writerow([eid, name, dept, des, sal])

print("Employee records saved successfully.")
```

```
... Employee ID: 12
    Employee Name: AJMAL
    Department: IT
    Designation: BACHELOR OF TECH
    Salary: 50000
    Employee records saved successfully.
```

Task 3 – JSON File: Faculty Information System

Problem Statement

Create a Python program to collect faculty details and store them in a JSON file named faculty.json.

Each faculty record should contain:

- Faculty ID
- Faculty Name
- Department

- Designation
- Experience (Years)

Store details of at least 5 faculty members.

```
import json

faculty = []

for i in range(1):

    data = {

        "Faculty ID": input("Faculty ID:"),

        "Faculty Name": input("Faculty Name:"),

        "Department": input("Department:"),

        "Email": input("Email:")

    }

    faculty.append(data)

with open("Faculty.json","w")as file:

    json.dump(faculty,file,indent = 4)

print("Faculty data is saved")
```

```
Faculty ID:99
Faculty Name:JD
Department:PERSONALITY DEVELOPMENT
Email:johndurairaj@gmail.com
Faculty data is saved
```

Task 4 – JSON File: Product Inventory Management

Problem Statement

Develop a Python program to collect product information and store it in a JSON file named products.json.

Each product should contain:

- Product ID
- Product Name
- Category
- Price
- Quantity Available

Store details of at least 10 products.

```
• Product ID: 89
  Product Name: GHEE
  Category: MILK
  Price: 150
  Quantity Available: 8
  Product data saved successfully.
```

Task 5 – XML File: Course Information Database

Problem Statement

Create a Python program to collect course details and store them in an XML file named courses.xml.

Each course should contain:

- Course Code
- Course Name
- Credits
- Department
- Faculty Name

Store details of at least 5 courses.

```
import xml.etree.ElementTree as ET
```

```
root = ET.Element("Courses")
```

```
for i in range(1):
```

```
    course = ET.SubElement(root, "Course")
```

```
    ET.SubElement(course, "CourseCode").text = input("Course Code: ")
```

```
    ET.SubElement(course, "CourseName").text = input("Course Name: ")
```

```
    ET.SubElement(course, "Credits").text = input("Credits: ")
```

```
    ET.SubElement(course, "Department").text = input("Department: ")
```

```
    ET.SubElement(course, "FacultyName").text = input("Faculty Name: ")
```

```
tree = ET.ElementTree(root)
```

```
tree.write("courses.xml")
```

```
print("Course data saved successfully.")
```

```
Course Code: CSA09
Course Name: JAVA
Credits: 10
Department: CSE
Faculty Name: MANIKAVELAN
Course data saved successfully.
```

Task 6 – XML File: Hospital Patient Management System Problem

Statement Develop a Python program to collect patient information and store it in an XML file named patients.xml. Each patient record should contain: ☐ Patient ID ☐ Patient Name ☐ Age ☐ Disease ☐ Doctor Assigned Store details of at least 10 patients.

```
import xml.etree.ElementTree as ET

root = ET.Element("Patients")

for i in range(1):

    patient = ET.SubElement(root, "Patient")

    ET.SubElement(patient, "PatientID").text = input("Patient ID: ")
    ET.SubElement(patient, "PatientName").text = input("Patient Name: ")
    ET.SubElement(patient, "Age").text = input("Age: ")
    ET.SubElement(patient, "Disease").text = input("Disease: ")
    ET.SubElement(patient, "DoctorAssigned").text = input("Doctor Assigned: ")

tree = ET.ElementTree(root)

tree.write("patients.xml")

print("Patient data saved successfully.")
```

```
Patient ID: 1001
Patient Name: sabarish
Age: 20
Disease: cancer
Doctor Assigned: no
Patient data saved successfully.
```